

Surrogate-Assisted Local Conditional Distribution Inference with Scarce Reference Outcomes

Abstract

Reference outcomes are costly, while cheaper surrogates are available for most units. We study the conditional distribution of a reference outcome at a chosen covariate value under selective verification, where few verified observations may lie nearby. A surrogate-free X -based estimator remains valid but can be noisy; surrogate outcome regression improves precision but can increase error when its local correction is inaccurate. We retain this surrogate-free estimator as an anchor and select one correction weight across the CDF by an observable local variance criterion. Sample splitting separates regression fitting and weight selection from final CDF evaluation. We establish path-oracle equivalence, a uniform Gaussian limit, and multiplier validity when the verification fraction may decrease. The bands cover a kernel-localized CDF; point-target CDF and quantile inference additionally require localization-bias control. Simulations show that the criterion distinguishes helpful from harmful full borrowing: the adaptive weight approaches zero under local reversal and retains gains under alignment, although Full can be more efficient under strong alignment. Two real-data applications and a controlled review benchmark show that the precision gain varies with the strength and direction of the local surrogate relation.

Keywords: two-phase sampling; missing at random; negative transfer; selective verification; semiparametric efficiency; multiplier bootstrap.

1 Introduction

Low-cost air sensors record particulate matter almost continuously, whereas regulatory or research-grade instruments operate at selected sites and times. For exposure assessment, the distribution of the reference measurement at a given humidity can be more informative than its conditional mean: upper-tail probabilities and quantiles describe episodes that an average conceals. Sensor readings are plentiful, but a calibration relation that is adequate overall may be inaccurate at that humidity or in that tail.

Pulse oximetry, expert review, and delayed-outcome studies create the same statistical problem. Covariates X and a surrogate S are observed for most units, while the reference outcome Y is observed only in a verified subset. We seek the conditional distribution of Y at $X = x_0$. The effective reference sample is therefore the verified sample near x_0 , which may be small even in a large study.

Verified outcomes support a natural estimator that uses an X -based outcome regression but leaves the surrogate out of the augmentation. We use this estimator as the anchor. Adding S to the regression can lower variance when it explains the threshold indicators for Y . The same addition can increase variance when the fitted surrogate correction is poor in the target neighborhood. The relevant comparison is the local variance change induced by the correction. A useful method must estimate how much of the correction to retain for the distribution at x_0 .

The CDF target makes this comparison more demanding than a scalar estimation problem. Threshold-by-threshold tuning can produce unrelated borrowing decisions along the same distribution. Tuning on the observations used to evaluate the final CDF also complicates a simultaneous confidence band. We therefore need one borrowing decision across the threshold range and an evaluation sample that has played no part in making that decision.

Validation-sample and two-phase methods recover information about variables measured in a smaller second-phase sample (Pepe, 1992; Tao et al., 2017). Augmented inverse-probability methods provide the corresponding missing-data representation and efficiency benchmark (Robins et al., 1994; Tsiatis, 2006; Cao et al., 2009). The data layout can also be viewed as semi-supervised: verified Y values act as labels, and the remaining units provide X and S . Much of the earlier semi-

supervised and prediction-powered literature targets prediction or a finite-dimensional population parameter (Chakraborty and Cai, 2018; Zhang et al., 2019; Azriel et al., 2022; Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024; Li and Ignatiadis, 2025; Miao et al., 2025). General semi-supervised theory already shows how auxiliary information can be weighted without imposing full borrowing. Song et al. (2024) derive projection-based optimal weights for general M-estimation and cross-validated inference, while Deng et al. (2024) construct optimal and safe estimators for high-dimensional regression under possible model misspecification. Chen et al. (2025) combine surrogate augmentation with regularized weighting and adaptive validation sampling for finite-dimensional estimating equations. Related work examines surrogate validity and efficiency in other inferential settings (Ji et al., 2025; Kallus and Mao, 2025). These procedures choose or regularize auxiliary-information use for parameter estimation rather than for a threshold-indexed local distribution.

Sui et al. (2026) localize prediction-powered inference for scalar conditional functionals at x_0 . Takeishi et al. (2026) use convolution smoothing to develop prediction-powered inference for finite-dimensional linear quantile-regression coefficients under possible misspecification, using separate labeled and unlabeled samples with a widely observed surrogate. Testa et al. (2025) allow missing-at-random labels and decreasing overlap for general smooth semiparametric targets. Wang (2026) develops uniform inference for transported counterfactual CDFs and quantiles with surrogate-assisted missing primary outcomes, whereas local regression-distribution methods assume that the outcome is fully observed (Cattaneo et al., 2024b,a). Here X and S are broadly observed, Y is selectively verified with p_M possibly tending to zero, and the target is the full process $F_0(\cdot \mid x_0)$. The effective reference information is $M p_M h^d$; the adaptive decision is one weight on an anchor-to-surrogate correction shared across thresholds; and the output comprises a uniform CDF process, simultaneous bands, and conditional quantile inversion. This combination creates a local, target-specific negative-transfer problem.

Finite-dimensional optimal or safe weights do not supply a shared path-oracle result when localization and verification both vanish, nor do scalar conditional procedures provide one borrowing

decision across a CDF followed by a post-selection simultaneous band. Theorems 1–3 establish these results for the present data structure.

Our construction uses a property of augmented inverse-probability estimation: with the correct verification probability, the outcome regression affects variance without changing the target. We fit an anchor regression using X and a second regression using (X, S) . Their difference defines a candidate surrogate correction. One weight, shared across thresholds, is selected by an observable local variance criterion. A weight of zero gives the anchor and a weight of one gives full borrowing. Separate samples fit the regressions, select the weight, and evaluate the CDF; rotating these roles lets every observation contribute to evaluation.

We introduce an adaptive borrowing rule for a local reference-outcome CDF: a single estimated weight governs the surrogate correction across thresholds while evaluation remains independent of fitting and selection. The theory establishes path-oracle equivalence, a uniform Gaussian limit and multiplier bands when the verification fraction may decrease, together with bias conditions for point-target CDF and quantile inference. Simulations test the transfer criterion under aligned and locally reversed surrogate relations, and the EPA, VitalDB, and AI-assisted review studies examine local distribution recovery under controlled reference sampling. Section 2 develops the estimator, Section 3 gives the large-sample results, Section 4 reports the simulations, Section 5 presents two empirical applications and a controlled review benchmark, and Section 6 concludes.

2 Surrogate-assisted local CDF estimation

2.1 Target and reference-outcome benchmark

We observe independent copies of $O = (W, R, RY)$, where W contains X, S , and any other broadly observed variables. We assume $R \perp Y \mid W$. In the primary designs $W = (X, S)$; an application may also include an observed variable used only by the verification design. The main procedure treats the verification probability $\pi_M(W)$ as known, as in a designed validation sample; the Supplement covers estimated probabilities. The reference outcome Y defines the estimand. The surrogate S enters only

through an augmentation used to improve precision. For $Z_y = 1(Y \leq y)$, let $q_{0,y}(X) = E(Z_y | X)$ and $m_{0,y}(W) = E(Z_y | W)$. At an interior point x_0 , we distinguish the scientific target from the target at the chosen localization scale:

$$\begin{aligned} F_0(y | x_0) &= \Pr(Y \leq y | X = x_0), \\ F_{h,0}(y | x_0) &= \frac{E\{K_h(X - x_0)Z_y\}}{\mu_h}, \quad \mu_h = E\{K_h(X - x_0)\}, \end{aligned} \tag{1}$$

where $K_h(X - x_0) = K\{(X - x_0)/h\}$. For a declared bandwidth, our confidence bands cover the finite-resolution target $F_{h,0}$. A symmetric second-order kernel gives a uniform $O(h^2)$ difference from F_0 under the smoothness conditions in the Supplement.

Any threshold-indexed augmentation $a_y(W)$ gives the unbiased signal

$$U_y(a) = a_y(W) + \frac{R}{\pi_M(W)}\{Z_y - a_y(W)\}. \tag{2}$$

The signal satisfies $E\{U_y(a) | W\} = m_{0,y}(W)$ and $E\{U_y(a) | X\} = q_{0,y}(X)$ for every a . Setting $a \equiv 0$ gives inverse-probability weighting. We call an X -based threshold regression the reference-outcome anchor because it can be fitted with the verified outcomes without using S in the augmentation. A regression using (X, S) gives full surrogate borrowing.

The augmentation changes variance without changing the target. Write $C_{yz}(W) = \text{Cov}(Z_y, Z_z | W)$ and $\delta_y^a(W) = a_y(W) - m_{0,y}(W)$. Then

$$\text{Cov}\{U_y(a), U_z(a) | W\} = \frac{C_{yz}(W)}{\pi_M(W)} + \left\{ \frac{1}{\pi_M(W)} - 1 \right\} \delta_y^a(W) \delta_z^a(W).$$

The first term is verified-outcome noise. The second is a positive-semidefinite covariance contribution from augmentation error. Thus a surrogate can predict Y well overall and still fail to reduce variance near x_0 . This identity motivates an adaptive amount of borrowing.

2.2 Adaptive surrogate borrowing

Condition on the regression-fitting sample and suppress hats and fold indices. Let $q_y(X)$ be the anchor candidate and let $m_y(W)$ use the surrogate. Their difference $D_y = m_y - q_y$ is the proposed correction; the unknown ideal correction is $A_y = m_{0,y} - q_y$. We use $a_{\lambda,y} = q_y + \lambda D_y$ with the same λ at every threshold. The measure ν specifies which thresholds matter for selection. We use equal mass on a fixed grid in the primary analyses, although a tail-focused analysis could place more mass in that region. For finite positive ν , the integrated augmentation variance that depends on λ is

$$\begin{aligned} J_{M,h}(\lambda) &= E\{v_{M,h}\|A - \lambda D\|_\nu^2\} \\ &= J_{M,h}(0) - 2\lambda G_{M,h} + \lambda^2 H_{M,h}, \\ G_{M,h} &= E\{v_{M,h}\langle A, D \rangle_\nu\}, \quad H_{M,h} = E\{v_{M,h}\|D\|_\nu^2\}, \end{aligned}$$

where

$$v_{M,h}(W) = \frac{h^d K_h^2(X - x_0)}{\mu_h^2} \left\{ \frac{1}{\pi_M(W)} - 1 \right\}.$$

Here $H_{M,h}$ measures the local size of the proposed correction and $G_{M,h}$ measures its alignment with the ideal correction. Full borrowing reduces integrated augmentation variance exactly when $2G_{M,h} - H_{M,h} > 0$.

When $H_{M,h} > 0$, the population choice is

$$\lambda_h^{\text{or}} = \Pi_{[0,1]} \left(\frac{G_{M,h}}{H_{M,h}} \right), \quad (3)$$

with $\lambda_h^{\text{or}} = 0$ when $H_{M,h} = 0$. The dependence on M is implicit. Missing-at-random verification gives an observable representation of $G_{M,h}$, even though A is unknown:

$$G_{M,h} = E \left[v_{M,h} \left\langle D, \frac{R}{\pi_M}(Z - q) \right\rangle_\nu \right].$$

One common weight need not minimize variance at each threshold. We therefore report lower- and upper-tail errors as well as integrated CDF error.

2.3 Cross-fitting and simultaneous inference

We keep weight selection separate from CDF evaluation. The sample is divided into three folds: one fits the two regressions, another selects the weight, and the third estimates the CDF. We rotate these assignments so that each observation appears once in an evaluation fold. In rotation k , the fitted functions $\widehat{q}_k$ and $\widehat{m}_k$ are fixed before the selection fold is used.

Let $\mathbb{P}_{G,k}$ denote the selection-fold empirical mean and $\widehat{\mu}_{h,G,k} = \mathbb{P}_{G,k} K_h(X - x_0)$. The empirical transfer weight is

$$\widehat{v}_{h,k}(W) = \frac{h^d K_h^2(X - x_0)}{\widehat{\mu}_{h,G,k}^2} \left\{ \frac{1}{\pi_M(W)} - 1 \right\}. \quad (4)$$

With $\widehat{D}_k = \widehat{m}_k - \widehat{q}_k$, the selection fold computes

$$\begin{aligned} \widehat{G}_{h,k} &= \mathbb{P}_{G,k} \left[\widehat{v}_{h,k} \left\langle \widehat{D}_k, \frac{R}{\pi_M} (Z - \widehat{q}_k) \right\rangle_v \right], \\ \widehat{H}_{h,k} &= \mathbb{P}_{G,k} \{ \widehat{v}_{h,k} \|\widehat{D}_k\|_v^2 \}, \\ \widehat{\lambda}_{h,k} &= \Pi_{[0,1]} \left\{ \frac{\widehat{G}_{h,k}}{\max(\widehat{H}_{h,k}, \varepsilon_{H,M})} \right\}, \end{aligned}$$

where $\varepsilon_{H,M} \downarrow 0$.

On the evaluation fold, use the selected path

$$\widehat{a}_{k,y} = \widehat{q}_{k,y} + \widehat{\lambda}_{h,k} (\widehat{m}_{k,y} - \widehat{q}_{k,y})$$

inside the augmented signal. Let $\widehat{F}_{h,k}$ be the corresponding kernel ratio and let $\rho_{M,k}$ be the evaluation-fold sample fraction. The combined raw estimator is

$$\widehat{F}_h^{\text{raw}}(y \mid x_0) = \sum_{k=1}^3 \rho_{M,k} \widehat{F}_{h,k}(y \mid x_0). \quad (5)$$

Conditional on the other two folds, the fitted functions and selected weight are fixed when the evaluation-fold mean is formed. This separation is the basis for the process limit and multiplier approximation developed below.

Algorithm 1: Local CDF estimation with adaptive surrogate borrowing

1. Fix x_0 , h , the threshold grid, ν , and three folds.
 2. Fit the X -based and surrogate-assisted threshold regressions on the fitting fold.
 3. Estimate $\widehat{G}_{h,k}$, $\widehat{H}_{h,k}$, and $\widehat{\lambda}_{h,k}$ on the selection fold.
 4. Estimate the augmented local CDF on the evaluation fold.
 5. Rotate the fold assignments and combine the evaluation-fold estimators using (5).
 6. Construct multiplier critical values from the evaluation-fold influence functions, and rearrange the point estimate for reporting and quantile inversion.
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The raw estimate can leave $[0, 1]$ or violate monotonicity. We report

$$\widehat{F}_h^{\text{mon}} = \mathcal{R}\{0 \vee \widehat{F}_h^{\text{raw}} \wedge 1\}, \quad \widehat{Q}_h(\tau \mid x_0) = \inf\{y : \widehat{F}_h^{\text{mon}}(y \mid x_0) \geq \tau\},$$

where $\mathcal{R}$ is increasing rearrangement. Inference for $F_{h,0}$ treats h as the inferential resolution and uses the raw-process band in Section 3. Inference for F_0 also requires undersmoothing or bias correction.

3 Large-sample theory for the local CDF

For a fixed bandwidth, the inferential target is the kernel-localized distribution $F_{h,0}(\cdot \mid x_0)$; Section 5 uses study-specific controlled-sampling benchmarks. We establish that estimating the borrowing weight has no first-order cost, derive a uniform CDF limit and a multiplier approximation, and then characterize the efficiency of surrogate borrowing. Replacing $F_{h,0}$ by the point target $F_0(\cdot \mid x_0)$ requires localization bias to be negligible at the stochastic scale.

3.1 Conditions and local information

Write

$$N_{M,h} = Mp_M h^d, \quad \kappa_K = \frac{\int K^2(u) du}{\{\int K(u) du\}^2}.$$

The quantity $N_{M,h}$ is the expected order of the verified sample in the target neighborhood. It, rather than M , determines the stochastic rate.

For a threshold-indexed function b , define the local verification metric

$$\|b\|_{\text{ver}}^2 = \sup_{y \in \mathcal{Y}} \frac{p_M h^d}{\mu_h^2} E \left[K_h^2(X - x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\} b_y^2(W) \right].$$

Assumption 1 (Observed-data identification and verification). Within each row M , the observations are independent and identically distributed, the joint law of (W, Y) is fixed, and $R \perp Y \mid W$. The known verification probability satisfies $\pi_M(W) = p_M e_M(W)$, where $p_M = E\{\pi_M(W)\} \rightarrow p \in [0, 1]$, $E\{e_M(W)\} = 1$, and $0 < c_e \leq e_M(W) \leq C_e < \infty$ almost surely.

Assumption 2 (Localization and the local-information rate). The point x_0 is interior to the support of X , and f_X is positive and continuous there. The nonnegative kernel is bounded, symmetric, compactly supported, and of bounded variation, with positive finite first and second integrals. The bandwidth satisfies $h \rightarrow 0$ and $N_{M,h} \rightarrow \infty$.

Assumption 3 (Candidate paths and sample separation). Regression fitting, weight selection, and CDF evaluation use disjoint samples whose limiting proportions are positive. The fitted threshold regressions are uniformly bounded and converge in the local verification metric to deterministic triangular paths q_M and m_M . The fold-specific oracle paths converge in that metric to a common population-minimizing path a_M , and a_M converges locally and uniformly over thresholds in conditional L_2 to a bounded path a_∞ .

Assumption 4 (Transfer-path curvature and degeneracy). For each rotation, the normalized curvature $\mathcal{H}_{h,k} = p_M H_{M,h,k}$ converges to $H_{\infty,k} \in [0, \infty)$. The stabilizer satisfies $\varepsilon_{H,M} \downarrow 0$ and $p_M \varepsilon_{H,M} = o(H_{\infty,k})$ when $H_{\infty,k} > 0$. Write $\mathcal{H}_{M,h}$ for the analogous normalized curvature of the deterministic

difference $m_M - q_M$. Then

$$\|m_M - q_M\|_{\text{ver}}^2 \leq \omega(\mathcal{H}_{M,h}) + o(1),$$

where ω is nondecreasing, continuous at zero, and $\omega(0) = 0$. The same relation holds in probability after replacing the deterministic difference and curvature by $\widehat{m}_k - \widehat{q}_k$ and $\mathcal{H}_{h,k}$, respectively.

Assumption 5 (Threshold-process regularity). (a) The threshold set is compact. Conditional CDF increments are locally Lipschitz in the threshold, and augmentation-path increments are locally Hölder in conditional L_2 . The path-oracle and fitted-difference classes are uniformly VC-type, have square-integrable envelopes satisfying the triangular-array Lindeberg condition, and have fitted-difference envelopes that vanish in the local L_2 metric.

(b) The local covariance integrand converges uniformly to a continuous kernel Γ_p defining a tight Gaussian process. For simultaneous bands, the distribution of $\|\mathbb{G}_p\|_{\mathcal{Y}}$ is continuous at the target critical value.

Here $e_0(W)$ denotes the local limit of $e_M(W)$ entering the covariance integrand in Assumption 5(b).

Assumption 6 (Point-target and quantile regularity). Uniformly over thresholds, the target CDF and covariate density have bounded continuous second derivatives in x near x_0 . For quantile inference, $F_0(\cdot \mid x_0)$ is continuously differentiable and strictly increasing on the threshold region, its values remain in an interior probability range, and $f_{Y|X}(\cdot \mid x_0)$ is bounded above and away from zero there.

Supplementary Section S1 gives the primitive conditions and their exact map to Assumptions 1–6. Write $\Delta_y(W) = a_{\infty,y}(W) - m_{0,y}(W)$.

3.2 Adaptive estimation of the local CDF

We first compare the feasible estimator with a path oracle. Let $\widehat{F}_h^{\text{or}}$ denote the sample-split estimator that uses the same fitted candidate regressions as the feasible estimator and replaces each selected

weight by its fold-specific population minimizer.

Theorem 1 (Adaptive path-oracle equivalence). *Under Assumptions 1–4 and Assumption 5(a), the criterion regret after multiplication by p_M is $O_{\mathbb{P}}(N_{M,h}^{-1/2} + \varepsilon_M)$, where $\varepsilon_M = p_M \varepsilon_{H,M}$. If the limiting curvature is positive, $\widehat{\lambda}_{h,k} - \lambda_{h,k}^{\text{or}} = O_{\mathbb{P}}(N_{M,h}^{-1/2})$. If, eventually, $\lambda_{h,k}^{\text{or}} \in [\delta, 1 - \delta]$ for some fixed $\delta > 0$, criterion regret is $O_{\mathbb{P}}(N_{M,h}^{-1})$. If curvature tends to zero, the fitted correction vanishes in the local process metric. In both cases,*

$$\sqrt{N_{M,h}} \left\| \widehat{F}_h^{\text{raw}} - \widehat{F}_h^{\text{or}} \right\|_{\mathcal{Y}} = o_{\mathbb{P}}(1).$$

The theorem separates adaptation from CDF estimation. The selection step is first-order negligible whether the candidate path has positive curvature or collapses to the anchor. The remaining stochastic rate depends on local verified information even when the overall verification fraction decreases. Define

$$\mathbb{Z}_M(y) = \sqrt{N_{M,h}} \{ \widehat{F}_h^{\text{raw}}(y \mid x_0) - F_{h,0}(y \mid x_0) \}, \quad (6)$$

$$\mathbb{Z}_M \Rightarrow \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}). \quad (7)$$

Theorem 2 (Uniform convergence of the kernel-localized CDF). *Under Assumptions 1–5, (7) holds for a tight mean-zero Gaussian process with covariance kernel*

$$\begin{aligned} \Gamma_p(y, z) = \frac{\kappa_K}{f_X(x_0)} E \left[\frac{C_{yz}(W)}{e_0(W)} + \left\{ \frac{1}{e_0(W)} - p \right\} \Delta_y(W) \Delta_z(W) \right. \\ \left. + p \{ m_{0,y}(W) - F_0(y \mid x_0) \} \{ m_{0,z}(W) - F_0(z \mid x_0) \} \mid X = x_0 \right]. \quad (8) \end{aligned}$$

The covariance identifies the part that adaptive borrowing can change: the term involving $\Delta_y \Delta_z$ is the cost of augmentation error. The remaining terms come from verified-outcome variation and heterogeneity within $X = x_0$; the latter vanishes when $p = 0$. Scarcity enters through p_M in the rate, while e_0^{-1} records where verification occurs locally. Equal verified counts can therefore yield

different precision at x_0 .

3.3 Simultaneous bands and conditional quantiles

Let $\mathbb{G}_M^\xi$ be the evaluation-fold Rademacher multiplier process formed from the fitted influence contributions. The Supplement gives its exact construction and the estimated-process replacement argument.

Theorem 3 (Multiplier process). *Under Assumptions 1–5,*

$$\mathbb{G}_M^\xi \Rightarrow_{\mathbb{P}} \mathbb{G}_p \quad \text{in } \ell^\infty(\mathcal{Y}). \quad (9)$$

If $c_{1-\alpha}^\xi$ is the conditional $(1 - \alpha)$ -quantile of the multiplier supremum, then

$$\Pr \left[F_{h,0}(y \mid x_0) \in \left\{ \widehat{F}_h^{\text{raw}}(y \mid x_0) \pm \frac{c_{1-\alpha}^\xi}{\sqrt{N_{M,h}}} \right\} \text{ for every } y \in \mathcal{Y} \right] \rightarrow 1 - \alpha. \quad (10)$$

One multiplier critical value therefore gives simultaneous coverage over the threshold region. This conclusion concerns $F_{h,0}$. The next result states the additional bias and shape conditions needed for the scientific point target and its quantiles.

Corollary 1 (Point-target CDF and conditional quantiles). *Suppose Theorems 2 and 3 and Assumption 6 hold. If $\sqrt{N_{M,h}}h^2 \rightarrow 0$, the process limit and confidence band remain valid with $F_0(\cdot \mid x_0)$ in place of $F_{h,0}(\cdot \mid x_0)$. If the point target is strictly increasing on the threshold region with conditional density bounded above and away from zero, increasing rearrangement is first-order neutral and, for compact $\mathcal{T}$ in the target probability range,*

$$\sqrt{N_{M,h}}\{\widehat{Q}_h(\cdot \mid x_0) - Q_0(\cdot \mid x_0)\} \Rightarrow \mathbb{Q}_p, \quad (11)$$

$$\mathbb{Q}_p(\tau) = -\frac{\mathbb{G}_p\{Q_0(\tau \mid x_0)\}}{f_{Y|X}\{Q_0(\tau \mid x_0) \mid x_0\}}. \quad (12)$$

The fixed-bandwidth simulations evaluate $F_{h,0}$. In Section 5, EPA and VitalDB use the full-data

empirical analogue $F_{h,\text{emp}}$, whereas the synthetic AI study uses its population target $F_{h,0}$. The feasible point-estimation experiments compare with F_0 .

3.4 Efficiency of surrogate borrowing

The canonical gradient identifies the efficiency benchmark, and the covariance identity below gives the exact cost of augmentation error.

Theorem 4 (Canonical gradient and local efficiency bound). *Fix M and h . Under Assumption 1, $\mu_h > 0$, and $E\{K_h^2(X - x_0)/\pi_M(W)\} < \infty$, the canonical gradient for the regularized target at threshold y is*

$$\phi_{M,h,y}^{\text{eff}}(O) = \frac{K_h(X - x_0)}{\mu_h} \left[m_{0,y}(W) + \frac{R}{\pi_M(W)} \{Z_y - m_{0,y}(W)\} - F_{h,0}(y \mid x_0) \right]. \quad (13)$$

Its covariance over any finite threshold collection is the semiparametric efficiency bound. The same bound applies when the verification law is unrestricted because the gradient is orthogonal to its tangent space.

This theorem establishes the efficiency benchmark for any feasible augmentation. The bound uses the true threshold regression $m_0(W)$. Orthogonality to the verification-law tangent space gives the same benchmark when a designed sampling probability is treated as known.

For augmentation a , let $\Omega_{M,h}^a(y, z)$ denote the covariance of the corresponding scaled influence functions.

Corollary 2 (Covariance cost of augmentation error). *Under Assumption 1, if the relevant second moments are finite, then*

$$\Omega_{M,h}^a(y, z) - \Omega_{M,h}^{m_0}(y, z) = \frac{h^d}{\mu_h^2} E \left[K_h^2(X - x_0) \left\{ \frac{1}{\pi_M(W)} - 1 \right\} \delta_y^a(W) \delta_z^a(W) \right]. \quad (14)$$

The matrix formed by this difference is positive semidefinite over every finite threshold set. Hence the oracle augmentation m_0 attains the efficiency bound, and augmentation error accounts exactly

for the covariance excess.

This identity turns surrogate quality into a covariance comparison at the target neighborhood. The adaptive path restricts that comparison to augmentations between the anchor and the fitted surrogate regression.

Theorem 5 (Local transfer criterion). *Under Assumption 1, conditional on the candidate regressions, suppose the weighted squares defining $J_{M,h}(0)$ and $H_{M,h}$ are integrable. The integrated augmentation variance is the quadratic criterion in Section 2. Its minimizer is λ_h^{or} when $H_{M,h} > 0$, and full borrowing improves on the anchor exactly when $2G_{M,h} - H_{M,h} > 0$. The path-oracle improvement is*

$$J_{M,h}(0) - J_{M,h}(\lambda_h^{\text{or}}) = \begin{cases} 0, & H_{M,h} = 0 \text{ or } G_{M,h} \leq 0, \\ G_{M,h}^2/H_{M,h}, & 0 < G_{M,h} < H_{M,h}, \\ 2G_{M,h} - H_{M,h}, & G_{M,h} \geq H_{M,h} > 0. \end{cases} \quad (15)$$

The criterion compares the anchor and Full along the fitted correction path. Its sign diagnoses full borrowing, while λ_h^{or} chooses the best point on that path. Neither statement implies finite-sample dominance over the anchor, a distinction examined in the numerical studies.

The Supplement gives complete proofs, the finite-threshold specialization, the limit of the efficiency bound, estimated-probability conditions, cluster inference, and local-linear extensions.

4 Simulation study

The simulations assess five aspects of the method: recovery of the conditional CDF, the transfer diagnostic, estimated borrowing under local misspecification, risk scaling with local verified information, and simultaneous coverage. Within each replication, all methods use the same observations, verification indicators, folds, and threshold grid. The Supplement reports the complete simulation designs, secondary learners, sensitivity analyses, and numerical results.

4.1 Design and estimands

The common design draws

$$X \sim U[-1, 1], \quad Y = \mu(X) + \sigma(X)\varepsilon,$$

where $\mu(x) = 1.2x + 0.6x^2$ and $\sigma(x) = 0.8 + 0.25x^2$. The baseline surrogate is $S = Y + \tau_S\eta$, with independent standardized errors. Most experiments target $x_0 = 0$ on a 101-point threshold grid corresponding to conditional probabilities from 0.02 to 0.98.

Mechanism and scarcity experiments use $h = 0.30$ and target $F_{h,0}$; feasible point-estimation experiments use $h = M^{-1/5}$ and target F_0 . Verification is random, bounded selective, or locally depleted near the target point, with probabilities calibrated to the stated mean reference fraction. Comparators are inverse-probability weighting, the X -based anchor, full surrogate borrowing, Adaptive, cross-validated shrinkage, and oracle variants.

The primary nuisance estimator is a location–scale model. Additional experiments use misspecified parametric and nearest-neighbor regressions, estimated verification probabilities, non-Gaussian and discrete outcomes, and two-dimensional localization.

Table 1: Finite-sample performance in the main simulation comparisons. ISE values are multiplied by 10^3 .

Contrast	Setting	Anchor	Full	Adaptive	Gain (%)	$E(\hat{\lambda}_h)$	NT Adaptive	NT Full
Reference fraction	$p_M = 0.05$	9.9	6.9	7.5	23.4	0.693	0.332	0.346
	$p_M = 0.10$	4.8	3.5	3.6	24.0	0.784	0.339	0.360
	$p_M = 0.20$	2.4	1.8	1.9	21.5	0.850	0.362	0.383
Surrogate noise	$\tau_S = 0.4$	4.9	2.3	2.5	48.5	0.835	0.203	0.222
	$\tau_S = 0.8$	4.7	3.5	3.6	23.7	0.787	0.337	0.370
	$\tau_S = 1.6$	4.7	4.3	4.4	7.6	0.683	0.429	0.443
Sample size	$M = 1,000$	8.2	6.3	6.6	19.1	0.713	0.379	0.391
	$M = 4,000$	2.7	2.0	2.1	25.3	0.835	0.350	0.361
Local reversal	N5	4.9	6.7	4.9	-0.1	0.002	0.015	0.823

Gain is the percentage reduction in mean ISE from the anchor. Full uses borrowing weight one. NT is the frequency with which a method has larger ISE than the anchor in the same replication. N5 is the local-reversal setting.

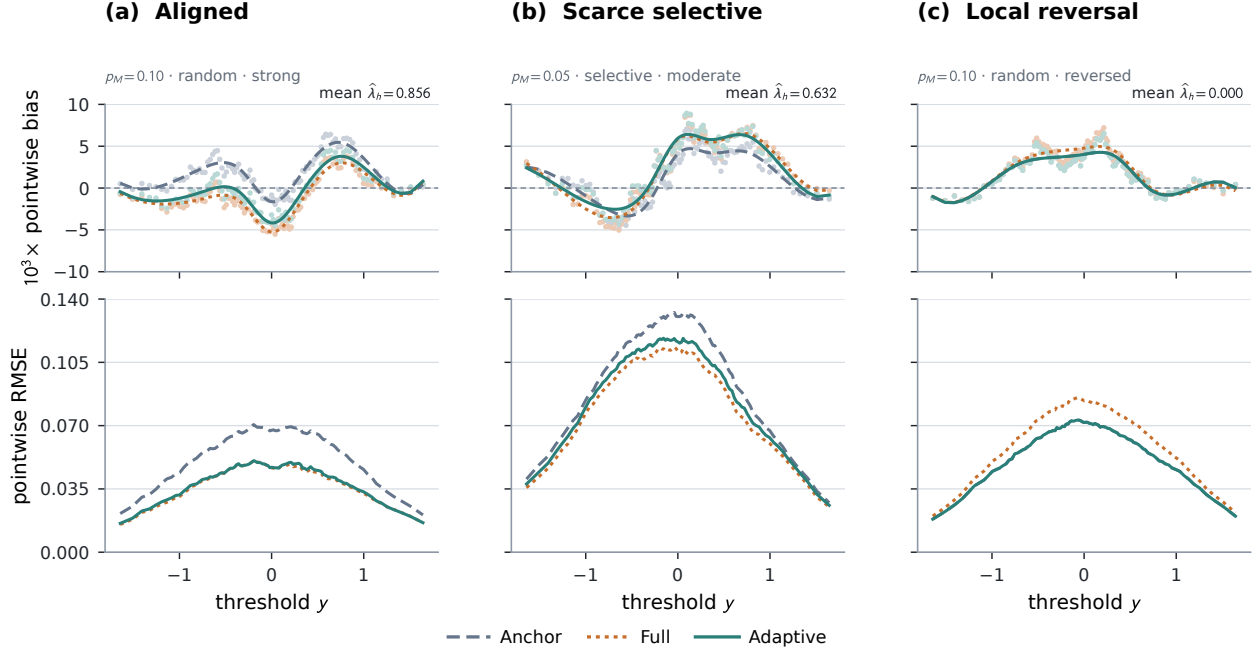


Figure 1: Pointwise recovery of the kernel-localized conditional CDF target $F_{h,0}(y | x_0)$ in three prespecified simulation settings. Columns show aligned borrowing, scarce selective verification, and local reversal. The upper row reports 1,000 times the Monte Carlo pointwise bias, defined as the mean rearranged estimate minus the target. Faint points are the unsmoothed biases; lines are inverse-Monte-Carlo-variance weighted cubic B-splines using the same five internal threshold knots for every curve. The horizontal reference is zero. The lower row reports pointwise RMSE on a common vertical scale. Each curve uses the same 500 Monte Carlo replications and 101-threshold grid. Full and Adaptive have lower RMSE than Anchor throughout the displayed grid in the aligned and scarce settings. Under local reversal, Adaptive tracks Anchor while Full has higher RMSE. No interval or confidence band is displayed.

4.2 Transfer direction and oracle-weight recovery

The oracle experiments vary the candidate correction while holding the reference-outcome distribution, localization, and threshold grid fixed. M1 and M2 favor full borrowing; M3 and M4 produce interior path-oracle weights; M5 and M6 reverse the correction globally or near x_0 ; and M7 and M8 introduce a lower-tail failure or smooth model error. Each path is evaluated under random and selective verification. The resulting 18 cells separate two questions: whether $2G_h - H_h$ predicts the risk direction of Full, and whether the estimated coefficient recovers the minimum along the candidate path.

Oracle augmentation lowers mean ISE relative to the X -based anchor by 25.3–28.0% in M1–M2.

The conservative M2 correction improves on the anchor without attaining the oracle- m risk, as predicted by the augmentation excess covariance. Across all 18 cells, the sign of $2G_h - H_h$ agrees with the paired anchor-minus-Full ISE direction under both verification designs.

The clipped and raw M4 paths isolate the effect of path consistency. Clipping changes the correction that enters the final estimator: the clipped path has a positive transfer signal and a positive realized gain, whereas the raw over-correction has a negative signal and increases ISE. Thus, the diagnostic tracks the correction actually used by the estimator, including transformations imposed before evaluation.

The estimated weight follows the same geometry. When $\lambda_h^{\text{or}} = 1$, its mean ranges from 0.852 to 0.998. The M3–M4 means remain close to their interior oracle values, and the lower-tail and smooth-error paths retain substantial but incomplete borrowing. In the global and local reversals, $\lambda_h^{\text{or}} = 0$, the largest mean estimate is below 10^{-4} , and Full increases risk. The mean absolute error between the 18 estimated and oracle cell means is 0.034.

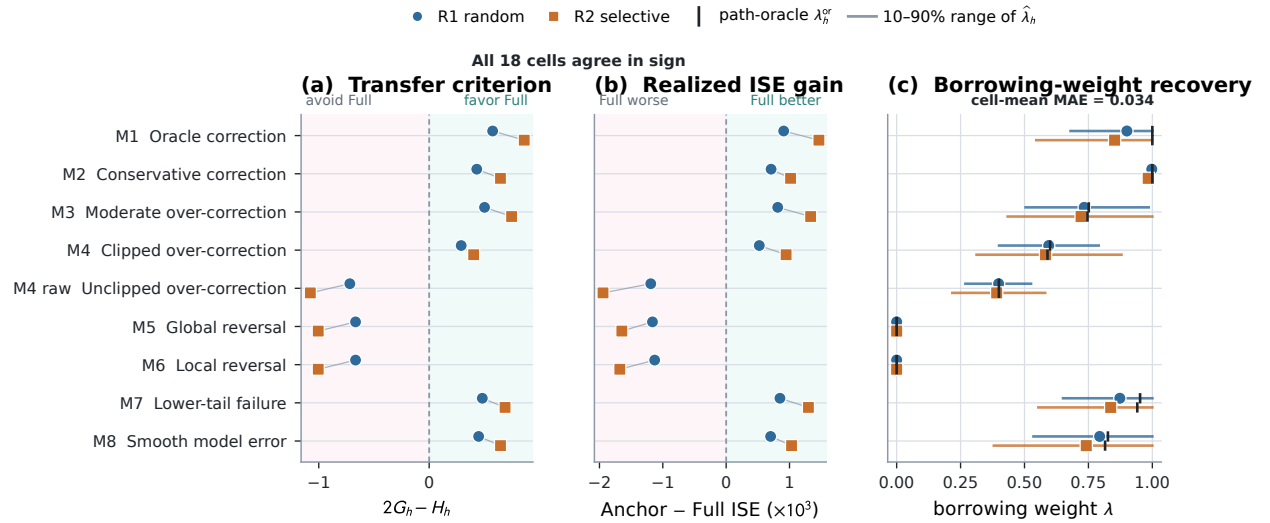


Figure 2: Transfer criterion, realized risk, and borrowing-weight recovery in the oracle experiments. Rows index the eight candidate mechanisms, with the clipped and raw M4 paths shown separately; circles denote R1 random verification and squares R2 selective verification. Panels (a) and (b) report the population criterion $2G_h - H_h$ and the paired Monte Carlo contrast $10^3 \{ \text{ISE}(\text{Anchor}) - \text{ISE}(\text{Full}) \}$. Positive values favor Full in both panels, and the signs agree in all 18 cells. Thin gray segments join the two verification designs for the same candidate path. Panel (c) shows the path-oracle weight (black tick), mean selected weight (colored marker), and empirical 10th–90th percentiles (colored segment; 2,000 replications for the primary paths and 1,000 for the raw M4 path); the mean absolute error across the 18 cell means is 0.034.

4.3 Adaptive borrowing under misspecification

The feasible experiments replace the oracle paths by regressions fitted within the three-way rotation. Along the reference-fraction sequence in Table 1, Adaptive reduces mean final-CDF ISE by 21.5–24.0%; absolute risk falls with p_M , while the mean weight rises from 0.693 to 0.850. Surrogate quality has a larger effect on the value of borrowing: the gain falls from 48.5% at $\tau_S = 0.4$ to 7.6% at $\tau_S = 1.6$, accompanied by a decrease in the mean weight from 0.835 to 0.683. Increasing M from 1,000 to 4,000 raises the gain from 19.1% to 25.3% and the mean weight from 0.713 to 0.835.

Full has the lowest mean ISE in these aligned cells because Adaptive estimates its coefficient on a separate selection fold. That gap contracts with local information. The Adaptive-minus-fitted-path-oracle ISE difference decreases from 7.17×10^{-4} to 3.69×10^{-5} as p_M rises from 0.05 to 0.20, and from 4.29×10^{-4} to 4.24×10^{-5} as M rises from 1,000 to 4,000. Adaptive has no higher negative-transfer frequency than Full in all 15 primary learner-by-quality cells and in seven of eight nuisance-stress cells.

The misspecified regression learners preserve the main risk pattern. At moderate surrogate noise, Adaptive gains are 23.7%, 25.3%, and 22.4% for the primary, misspecified parametric, and nearest-neighbor candidates. Over the strong-to-weak surrogate sequence, the corresponding gain ranges are 48.5–7.6%, 49.8–11.0%, and 37.0–8.0%. The three learners differ in how much of the available surrogate information they recover, but all preserve the declining gain as the surrogate becomes noisier.

In the local-reversal setting N5, Adaptive estimates a mean weight of 0.0017 and has a negative-transfer frequency of 1.5%, compared with 82.3% for Full. A small number of loss realizations leave its mean ISE 0.066% above the anchor.

The vanishing-path experiment addresses the weak-curvature case in Theorem 1. For path scales $s \in \{1, 0.5, 0.25, 0.125, 0\}$, relative curvature follows s^2 , the effective correction $sE(\widehat{\lambda}_h)$ decreases from 0.899 to zero, and the Adaptive-minus-path-oracle ISE gap decreases from 6.01×10^{-5} to zero. Coefficient identification weakens with the path, while the correction entering the CDF disappears at the same time.

4.4 Verification scarcity and simultaneous bands

For local-information rate indices $Mp_Mh \geq 30$, scaled ISE ranges from 0.319 to 0.358 for the anchor and from 0.241 to 0.286 for Adaptive. Selective verification changes the covariance constant even at the same nominal reference fraction, so local precision depends on verification allocation as well as the reference count.

The rescaled risks are stable over combinations of M and p_M that produce the same rate index. Above 30, their coefficients of variation are 0.032 for the anchor and 0.052 for Adaptive. At indices 7.5 and 15, the remaining variation is larger, consistent with the small local reference samples. The ordering of the two methods persists throughout the grid, while the separation between random and selective verification remains after rate normalization because inverse-probability weighting changes the limiting covariance constant.

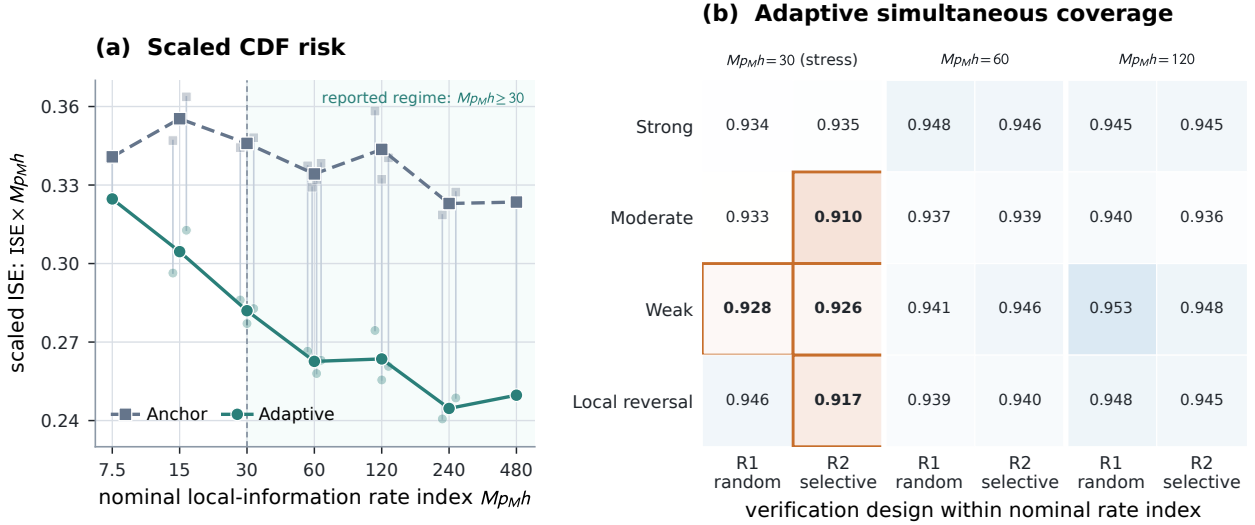


Figure 3: Risk scaling and raw-process simultaneous inference for the kernel-localized target $F_{h,0}$ under scarce reference outcomes. In panel (a), each thin vertical segment joins the Anchor and Adaptive mean scaled ISE from the same scarcity setting; the large points and lines are unweighted cell means at each nominal rate index. Each scenario uses 1,000 replications, and the shaded region marks the reported scaling regime $Mp_Mh \geq 30$; the spread across scenarios is not an uncertainty interval. Panel (b) reports Adaptive empirical coverage of the 95% raw-process simultaneous band over 1,000 replications and 4999 multiplier draws per replication. Cell color is centered at 0.95, and outlined bold cells are below 0.93; binomial Monte Carlo standard errors range from 0.0067 to 0.0090. Supplementary Table S4 gives Anchor coverage and integrated widths, and Table S5 gives the corresponding local-support diagnostic.

The band grid uses nominal rate indices $Mp_Mh = 30, 60, 120$; the first is a deliberately

low-information stress regime. In its selective-verification cells, Adaptive and the anchor both under-cover. The minimum Adaptive estimate, 0.910, has Monte Carlo standard error 0.0090. The minimum role-fold verified support averages about five in these cells. Pooled over the band grid, Adaptive coverage increases from 0.875 at counts 0–3 to 0.911 at counts 4–5, 0.943 at counts 6–9, and 0.945 when the count is at least 10. The minimum count across the fitting, selection, and evaluation roles is therefore more informative for finite-sample calibration than the global reference fraction alone.

Across all 24 cells, Adaptive coverage ranges from 0.910 to 0.953. Under selective verification, mean coverage increases from 0.922 at $p_M = 0.05$ to 0.944 at $p_M = 0.20$. Strong aligned surrogates reduce integrated width to 0.760–0.981 of the anchor width; local reversal produces ratios of 1.000–1.003 because the fitted path contributes almost no correction. A prespecified sensitivity increased endpoint rearrangement in seven cells and improved several low-information coverages without removing the common shortfall. Pointwise studentization reduced coverage to 0.358–0.633, so the raw-process band remained the primary procedure.

4.5 Robustness across outcomes and nuisance fits

The non-Gaussian experiments replace the normal error while retaining the same conditional mean, scale, and surrogate construction. Adaptive gains are 15.6%, 17.8%, and 24.7% under standardized t_3 , standardized χ_3^2 , and Gaussian-mixture errors, compared with 22.0% under normal errors. The method continues to borrow under heavy tails and skewness, but the gain changes with the thresholdwise regression difficulty.

For a ten-level ordered outcome, category-scale oracle truth gives a 22.5% Adaptive gain and a 25.9% Full gain. The target and risk are evaluated on the support of the discrete outcome, matching the finite-grid process result.

The two-dimensional experiments use $X = (X_1, X_2)$, a product Epanechnikov kernel, and $h = M^{-1/6}$. Across $M \in \{2,000, 5,000\}$ and $p_M \in \{0.10, 0.20\}$, Adaptive gains range from 20.9% to 24.4%. The larger neighborhoods required in two dimensions follow the effective information

scale $M p_M h^d$.

Eight exploratory configurations estimate the verification probability on the fitting fold. Adaptive gains range from 22.9% to 45.4%, with mean weights from 0.408 to 0.600. The evaluated designs retain a risk benefit after the verification model is estimated. Process inference with a learned verification model uses the local product-rate and drift conditions stated in the Supplement.

5 Empirical applications and a controlled review benchmark

The empirical evidence consists of two real-data applications and one controlled synthetic benchmark. The EPA study concerns a low-cost air sensor observed alongside a reference monitor; the VitalDB study pairs continuous pulse oximetry with intermittent arterial-blood-gas measurements; and the AI study pairs an automated score with expert review. These settings represent strong local surrogacy, weaker and target-dependent surrogacy, and a controlled local failure of an otherwise accurate surrogate. Table 2 records how the scientific variables map to (X, S, Y) .

The target in each study is a local reference-outcome CDF. The analysis starts from complete paired outcomes and then reveals Y under known reference-sampling probabilities while retaining (X, S) for every unit. For EPA and VitalDB, the benchmark $F_{h,\text{emp}}$ is the kernel-localized empirical CDF computed from all available reference outcomes. The synthetic AI benchmark instead uses the population target $F_{h,0}$. Repeated reference sampling measures recovery of these study-specific local distributions.

5.1 Air-quality distributions at fixed humidity

Dense low-cost sensor networks can provide much broader coverage than regulatory or research-grade monitors, but the field relationship between the two instruments changes with environmental conditions. Humidity is especially relevant to particle-sensor response. The resulting question is distributional: at a specified humidity, how much mass and upper-tail probability would the reference monitor assign to different $\text{PM}_{2.5}$ levels? A correction aimed only at the conditional mean

Table 2: Data structure and inferential role of the empirical studies.

Study	Analysis units	Reference outcome and surrogate	Localization	Distributional question
EPA air sensors	2,814 collocated site-days at seven U.S. sites	Reference-monitor $\text{PM}_{2.5}$; uncorrected PurpleAir $\text{PM}_{2.5}$	Relative humidity at 30%, 50%, and 70%	Recovery of humidity-specific $\text{PM}_{2.5}$ distributions and upper-tail quantiles from sparse reference monitoring
VitalDB oximetry	3,178 subjects with one matched measurement pair each	Arterial-blood-gas SaO_2 ; pulse-oximeter SpO_2	Elapsed case time at 30, 75, and 180 minutes	Recovery of a discrete saturation distribution when the continuous surrogate becomes less informative locally
AI-assisted review	2,998 evaluated dossiers in a controlled synthetic corpus	Mean of three expert ratings; automated score	Standardized complexity at $x_0 = 0$	Recovery of expert-score distributions under scarce or selective review, including controlled local surrogate reversal

Table 3: Primary results under controlled reference sampling. EPA and VitalDB use their localized complete-data empirical benchmarks; AI uses its population benchmark.

Study	Target	Ref. frac.	Anchor	Full	Adaptive	Mean wt.	Adaptive/Anchor [95% MC]
EPA	30% RH	10.0%	2.060	0.738	0.818	0.846	0.397 [0.337, 0.457]
	50% RH	10.0%	0.864	0.375	0.386	0.888	0.447 [0.385, 0.510]
	70% RH	10.0%	1.601	0.722	0.765	0.862	0.478 [0.423, 0.534]
VitalDB	30 min	10.0%	0.205	0.177	0.181	0.803	0.882 [0.851, 0.914]
	75 min	10.0%	0.393	0.347	0.349	0.867	0.888 [0.857, 0.919]
	180 min	10.0%	0.241	0.232	0.234	0.672	0.971 [0.949, 0.992]
AI	Random review	2.5%	9.091	3.380	4.195	0.775	0.461 [0.427, 0.496]
	Random review	10.0%	2.147	0.814	0.861	0.891	0.401 [0.363, 0.439]
	Random review	40.0%	0.434	0.197	0.199	0.955	0.457 [0.421, 0.494]

ISE values are multiplied by 10^3 . Risk-ratio intervals use common reference samples and quantify Monte Carlo variation conditional on the stated study-specific benchmark; they are not population-sampling confidence intervals.

does not determine that local distribution.

We use the public EPA long-term air-sensor study, in which six commercial sensor types were collocated with reference monitors at seven U.S. sites in 2019–2021 (Barkjohn et al., 2025; U.S. Environmental Protection Agency Office of Research and Development, 2025). Uncorrected PurpleAir hourly measurements are aggregated to 2,814 site-days. Set $Y = \log(1 + \text{reference } \text{PM}_{2.5})$, $S = \log(1 + \text{PurpleAir } \text{PM}_{2.5})$, and X to relative humidity; the targets are 30%, 50%, and 70%.

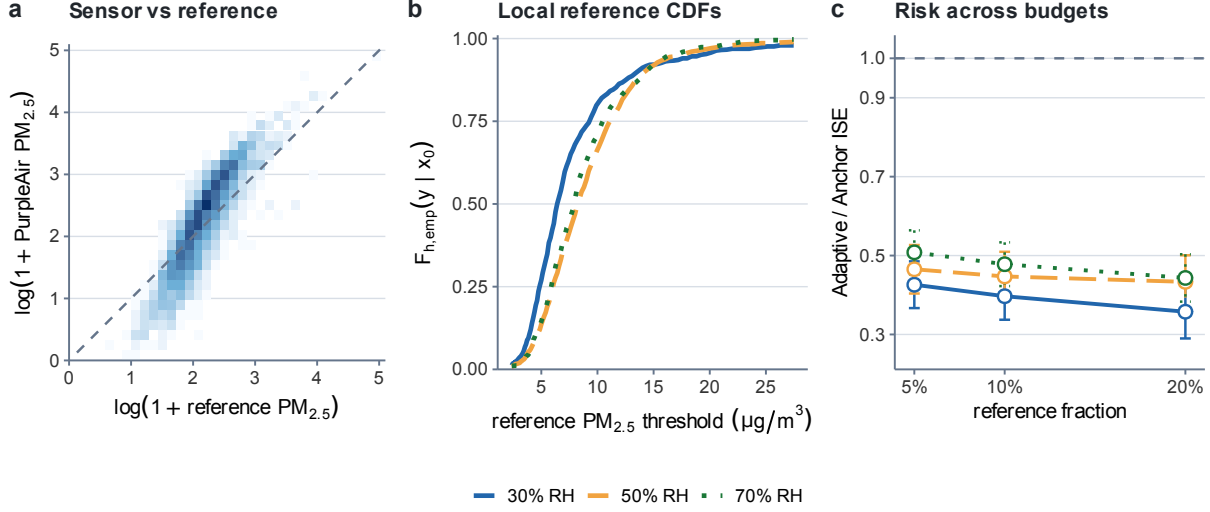


Figure 4: EPA long-term air-sensor application. Panel (a) displays all 2,814 collocated site-days on the log scale; darker bins contain more observations and the dashed line is equality between the PurpleAir and reference-monitor measurements. Panel (b) shows the complete-data kernel-localized empirical CDF at 30%, 50%, and 70% relative humidity, using a bandwidth of ten humidity percentage points. Panel (c) reports Adaptive-to-Anchor mean ISE ratios under 5%, 10%, and 20% random reference sampling. Vertical ranges are paired 95% Monte Carlo intervals from 500 reference-sampling repetitions; the horizontal line at one is the Anchor risk.

PurpleAir is an imperfect surrogate; the complete collocations define the reference distribution.

Random reference sampling and sampling selective on sensor level and humidity are examined at 5%, 10%, and 20% budgets. Calendar weeks, rather than individual site-days, are assigned to fitting, weight selection, and CDF evaluation, preserving the temporal grouping in the collocation study. Each design–budget–target combination is evaluated over 500 reference-sampling repetitions, with the same reference sample used for all methods.

Figure 4(a) shows a strong overall sensor–reference relationship together with appreciable spread around equality. The complete-data local CDFs in panel (b) also change across humidity targets; the 30% curve places more mass at lower $\text{PM}_{2.5}$ thresholds than the 50% and 70% curves. Thus the application involves both calibration error and a target distribution that varies with the localization variable.

At the 10% random-reference budget in Table 3, Adaptive reduces mean ISE by 52.2–60.3% across the humidity targets, with the largest gain at 30%. All three paired Monte Carlo intervals

for the risk ratio exclude one, and mean borrowing remains between 0.846 and 0.888. Full has lower mean ISE at all three targets, with paired intervals also favoring Full. Adaptive’s observed negative-transfer frequency ranges from 9.0% to 13.6%, compared with 10.0–15.8% for Full; the largest separation occurs at 50% humidity.

Panel (c) extends that comparison over all nine random-sampling cells. Adaptive-to-Anchor risk ratios range from 0.358 to 0.508, and every paired interval lies below one. The gain is largest near 30% humidity but remains substantial throughout the humidity and reference-budget ranges.

The gain extends to the upper tail: the conditional 0.90-quantile error is lower at every humidity target. At 30% humidity, Adaptive reduces mean absolute error from 2.00 to 0.83 micrograms per cubic meter; the reductions at 50% and 70% have the same direction. For the fixed reference sample, weekly cluster-multiplier bands are 19–30% narrower than the anchor bands and contain the complete-data CDF at all three targets.

5.2 Oxygen-saturation distributions during surgery

Pulse oximetry supplies a continuous, noninvasive oxygen-saturation signal, whereas arterial-blood-gas analysis supplies a more direct but intermittent reference measurement. Their relationship need not be constant over a surgical case. Moreover, the lower tail of the SaO_2 distribution carries information that is not summarized by a mean calibration equation, while the upper end is strongly discrete: saturation values are bounded and often recorded at 100.

VitalDB combines high-resolution intraoperative monitoring with laboratory measurements (Lee et al., 2022). We match each arterial SaO_2 measurement to the nearest pulse-oximeter SpO_2 reading from the same case within one second. An analysis-independent rule selects one pair per subject, leaving 3,178 independent subject-level records from 8,476 eligible pairs. Here Y is SaO_2 , S is SpO_2 , and $X = \log(1 + \text{elapsed case time})$. The local targets are 30, 75, and 180 minutes.

Random sampling and sampling favoring low SpO_2 values retain 5%, 10%, 20%, or 40% of arterial measurements as reference outcomes. Each design–budget combination is evaluated over 500 sampling repetitions; the reference samples are nested across budgets and shared across targets.

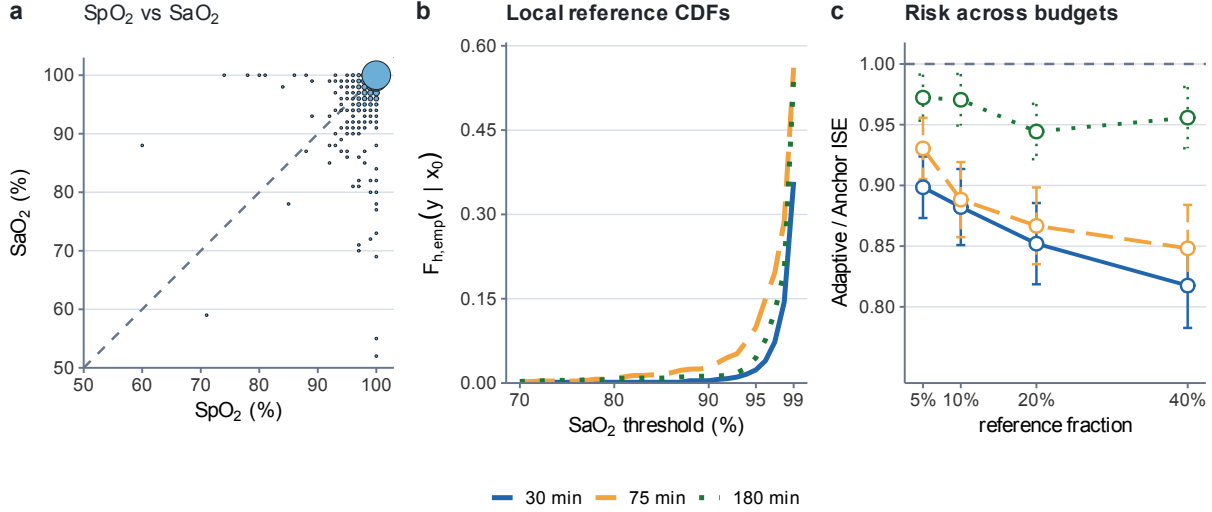


Figure 5: VitalDB oxygen-saturation application. Panel (a) displays all 3,178 subject-level SaO₂–SpO₂ pairs; point area is proportional to the number of observations at each integer pair and the dashed line is equality. Panel (b) shows the complete-data kernel-localized empirical SaO₂ CDF at 30, 75, and 180 minutes on the threshold grid from 70 through 99. Panel (c) reports Adaptive-to-Anchor mean ISE ratios under 5%, 10%, 20%, and 40% random reference sampling. Vertical ranges are paired 95% Monte Carlo intervals from 500 reference-sampling repetitions; the horizontal line at one is the Anchor risk.

Figure 5(a) displays the discreteness of both measurements and the concentration near the upper boundary. In the primary sample, 52.3% of SaO₂ values equal 100. Panel (b) shows that the local reference CDF also changes with elapsed time, particularly over the upper jumps represented on the threshold grid.

At the 10% random budget, Adaptive lowers mean ISE by 11.8%, 11.2%, and 2.95% at the three successive target times. All three paired anchor-minus-Adaptive intervals exclude zero. Full has lower mean ISE; its paired advantage over Adaptive is conclusive at 30 minutes but not at the two later targets.

The risk ratio reaches 0.971 at 180 minutes, and mean borrowing falls to 0.672 there from 0.803–0.867 at the earlier targets. Panel (c) shows the same ordering over the full budget grid: the largest gain occurs at 30 minutes, whereas the 180-minute ratios remain closest to one.

Adaptive’s observed negative-transfer frequency is 1.2–4.4 percentage points lower than Full’s; the paired frequency contrast excludes zero at 75 and 180 minutes. Adaptive nevertheless exceeds anchor risk in 35.4–46.8% of sampling repetitions, a larger frequency than in the EPA and aligned

AI studies.

The target-specific loss of surrogate information is also present in the paired data. Local SpO_2 – SaO_2 correlations are 0.498, 0.465, and 0.213 at the three target times. The 30-minute neighborhood contains only three values below 90, and the lower tail remains sparse at the later targets. The weaker 180-minute association and small lower-tail counts leave less scope for a surrogate correction even though the overall reference budget is unchanged.

Within the evaluated grid, the information multiplier for a 10% Adaptive budget decreases from 1.109 at 30 minutes to 1.027 at 180 minutes, mirroring the decline in local surrogate association.

5.3 Controlled benchmark for limited expert review

The third study is a controlled synthetic benchmark rather than an external data application. Automated assessment provides a broadly available surrogate, whereas panel review is costly enough to be collected only for a subset of cases. The target is the distribution of expert-panel scores at a fixed level of dossier complexity. This construction permits the review law and local surrogate error to be varied while retaining a known population benchmark—a combination unavailable in the two observational applications.

Here X is standardized complexity, S is the automated score, and Y is the mean of three expert ratings. The primary target is $x_0 = 0$, the center of the complexity scale. In confidence-triggered review designs, the broadly observed model-confidence score C also enters the known verification law through $W = (X, S, C)$; the augmentation candidates use (X, S) .

Synthetic dossiers contain performance dimensions, verified adverse events, operating complexity, and contextual information, but exclude the panel score and latent generating variables. Three independent corpora contain 3,000, 1,500, and 1,500 dossiers. Each dossier was scored under a common rubric before any expert-review samples were drawn; 5,997 of 6,000 evaluations yielded valid scores. Corpus A is used for the primary analysis, and a second automated evaluator provides a sensitivity analysis. The complete scoring specifications and prompts are reported in the Supplement.

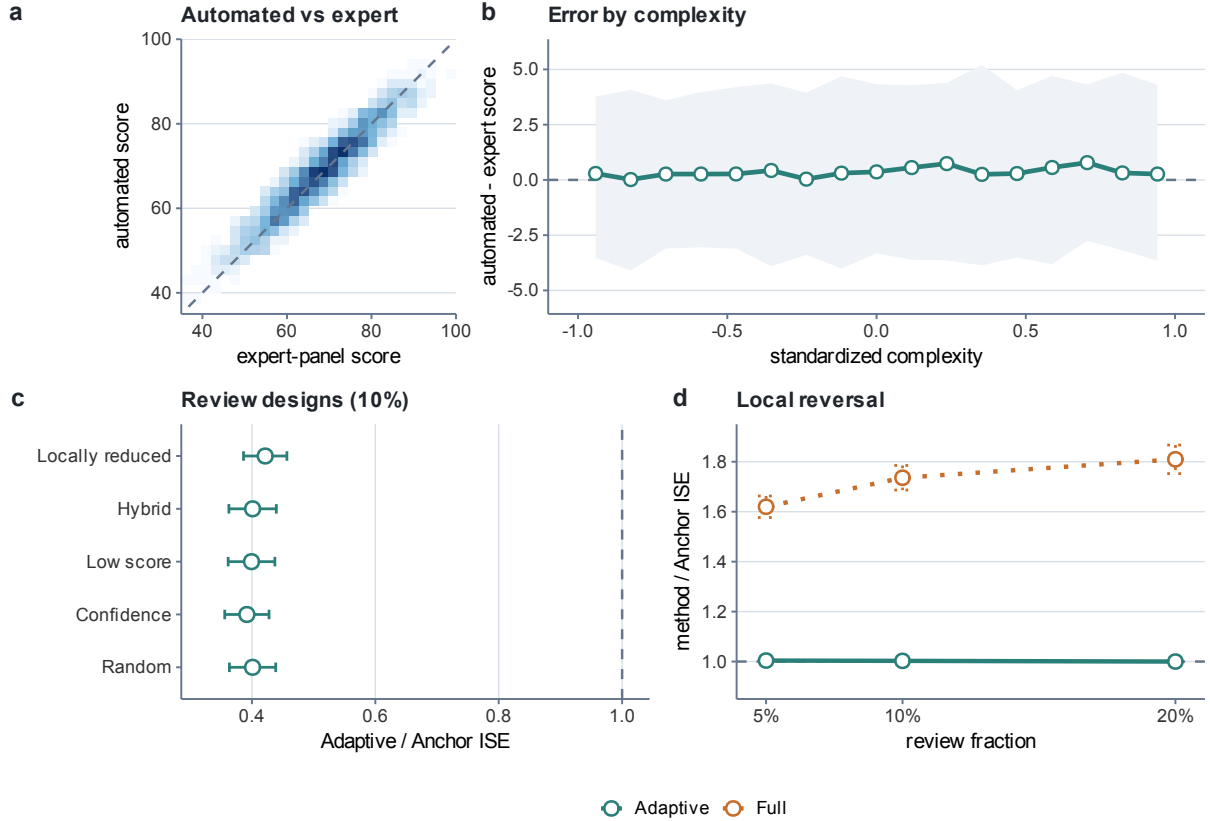


Figure 6: Controlled synthetic AI-assisted review benchmark. The risk benchmark is the population localized expert-score CDF at $x_0 = 0$. Panel (a) displays all 2,998 valid automated scores against corrected expert-panel scores; darker bins contain more dossiers and the dashed line is equality. Panel (b) reports mean automated-minus-expert score in 17 fixed complexity bins; the pale envelope spans the empirical 10th and 90th score-error percentiles within each bin. Panel (c) reports Adaptive-to-Anchor mean ISE ratios for five review designs at a 10% review fraction. Panel (d) reports Adaptive- and Full-to-Anchor mean ISE ratios under locally reduced review and local surrogate reversal. Intervals in panels (c)–(d) are paired 95% Monte Carlo intervals from 2,000 sampling repetitions; the horizontal line at one is the Anchor risk.

Each review design is evaluated over 2,000 sampling repetitions. The designs include random review, confidence- or score-triggered review, a hybrid rule, and reduced review near $x_0 = 0$. Each design is calibrated to its stated mean review fraction.

On the 2,998 valid Corpus-A cases, the AI score has mean absolute error 2.415 and Pearson correlation 0.946 with the corrected expert outcome; the calibration slope is 0.992. The maximum absolute local bias over the analysis bins is 0.782. Reported confidence has correlation 0.159 with negative absolute error. High global score correlation therefore coexists with local bias and a weak confidence-based review signal.

The overall score relation in Figure 6(a) is strong, but panel (b) shows that automated-minus-expert error still varies across local complexity bins. This separates global predictive accuracy from the local distributional relation on which borrowing at x_0 depends.

Under random review, Adaptive reduces mean ISE by 53.9–59.9% over reference fractions 0.025–0.40. Its mean borrowing weight increases from 0.775 to 0.955, while the Adaptive-to-anchor risk ratio remains between 0.401 and 0.461. Full has lower mean ISE at every fraction, although the absolute Full–Adaptive gap contracts as the review fraction increases. For fractions up to 0.10, the anchor requires 2.28–2.50 times the actual random-review fraction to match Adaptive risk within the evaluated grid.

Across the five primary review designs, Adaptive gains range from 35.8% to 61.1%. Confidence-triggered review closely follows random review, consistent with the weak confidence–error association. Reduced review near x_0 has the smallest gain at the lowest review budget; the selection and evaluation folds then contain fewer local reference observations. At the 10% budget, the five Adaptive-to-Anchor ratios in Figure 6(c) lie between 0.392 and 0.422.

With locally reduced review and surrogate reversal, the mean weight falls from 0.072 to 0.001 as p rises from 0.05 to 0.20; mean losses are 0.39%, 0.27%, and approximately zero. Full’s negative-transfer frequency is 75.3–79.4%.

In six representative AI band settings, Adaptive simultaneous coverage is 0.940–0.974. Its integrated bands are 6.8–29.7% narrower than anchor bands in every setting. Reduced local review produces the widest bands and the lowest Adaptive coverage, whereas the aligned random and hybrid designs retain the largest width reductions.

5.4 Comparison across the studies

Across the examined budget grids, Adaptive-to-anchor risk ratios are 0.358–0.508 for EPA, 0.401–0.461 for random AI review, and 0.818–0.972 for VitalDB. Every paired interval remains below one, but the size of the gain is target specific. EPA and the aligned AI design combine high borrowing with large risk reductions. VitalDB yields smaller gains, especially at 180 minutes where the local

SpO₂–SaO₂ relation is weakest.

Full often has slightly lower mean risk in the aligned settings. That ordering reflects the cost of estimating a weight on a separate fold when the best path is close to full borrowing. The comparison changes under local reversal: Adaptive selects weights near zero and sharply lowers the observed negative-transfer frequency relative to Full. Across the three studies, the selected weight follows local distributional alignment more closely than any single global measure of surrogate accuracy.

6 Discussion

The surrogate enters through augmentation and, with a correct verification probability, can change variance without changing the local reference-outcome distribution being estimated. A shared weight coordinates borrowing across thresholds, and separate evaluation folds support simultaneous inference.

The observed-data structure is semi-supervised: verified reference outcomes are labels, and the other units contribute covariates and surrogate measurements. The inferential task differs from ordinary semi-supervised prediction. We estimate a feature of the reference-outcome distribution under selective verification, and the surrogate regression is an efficiency device. The observable criterion $2G_h - H_h$ compares Full with the anchor along the fitted correction path at x_0 .

Full has lower mean ISE when the surrogate correction is well aligned. Adaptive loses some efficiency in those settings because it estimates the weight on a separate fold. Under local reversal, the selected weight is almost zero and the Adaptive CDF stays close to the anchor, while Full has larger error. If the candidate correction itself vanishes, its coefficient is weakly identified, but changing that coefficient has negligible effect on the estimated CDF.

The verified fraction and its local allocation both determine precision. In $\pi_M(W) = p_M e_M(W)$, p_M determines the effective local rate and e_M records how verification is distributed near the target. Studies with equal numbers of reference outcomes may therefore have different precision at x_0 . Simultaneous coverage remains below the nominal level when selective verification leaves very little

local information. The minimum verified kernel-support count across the three role folds is directly observable; the simulations flag counts below 10 as low information for simultaneous bands.

The fixed-bandwidth simulations and the synthetic AI study use $F_{h,0}$, the population distribution at the declared localization scale; the EPA and VitalDB reference-sampling studies evaluate the corresponding full-data empirical benchmark $F_{h,\text{emp}}$. Bands for F_0 require undersmoothing or bias correction. Kernel localization is most practical for a low-dimensional conditioning variable.

The reference outcomes define the distribution of interest; the local transfer criterion governs how the surrogate enters its estimation.

Supplementary Materials

The Supplement contains primitive conditions, complete proofs, finite- and growing-grid results, point-target and quantile theory, estimated-probability and local-linear extensions, independent-cluster inference, optimal local verification allocation, the adjacent-method comparison, and additional numerical and empirical details.

Data and Code Availability

The EPA air-sensor and VitalDB data are publicly available from the sources cited in Section 5. Code and derived numerical data sufficient to reproduce every table and figure accompany the submission. The complete computational materials also retain the synthetic dossiers, expert-panel outcomes, automated scores, and intermediate simulation results; these materials are available to the editors during review.

Disclosure Statement

The authors declare that there are no conflicts of interest.

Generative AI Use Statement

DeepSeek models `deepseek-v4-pro` and `deepseek-v4-flash` were used as automated surrogate evaluators in the controlled synthetic AI-assisted review benchmark. They scored the synthetic dossiers under a common rubric before the expert-review samples were drawn; expert-panel outcomes defined the reference estimand. OpenAI Codex (GPT-5) was used during manuscript preparation to assist with code debugging, selective editorial revision, language refinement, and checks of wording, consistency, citations, and typography. The authors reviewed and verified the research design, statistical analyses, all code changes, and all revised text. All authors approved these uses and the present disclosure and accept responsibility for the article.

References

- Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023a. doi: 10.1126/science.adi6000.
- Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*, 2023b. doi: 10.48550/arXiv.2311.01453.
- David Azriel, Lawrence D. Brown, Michael Sklar, Richard Berk, Andreas Buja, and Linda Zhao. Semi-supervised linear regression. *Journal of the American Statistical Association*, 117(540): 2238–2251, 2022. doi: 10.1080/01621459.2021.1915320.
- Karoline K. Barkjohn, Robert Yaga, Brittany Thomas, William Schoppman, Kenneth S. Docherty, and Andrea L. Clements. Evaluation of long-term performance of six PM2.5 sensor types. *Sensors*, 25(4):1265, 2025. doi: 10.3390/s25041265.
- Weihua Cao, Anastasios A. Tsiatis, and Marie Davidian. Improving efficiency and robustness of the doubly robust estimator for a population mean with incomplete data. *Biometrika*, 96(3):723–734, 2009. doi: 10.1093/biomet/asp033.

- Matias D. Cattaneo, Rajita Chandak, Michael Jansson, and Xinwei Ma. Boundary adaptive local polynomial conditional density estimators. *Bernoulli*, 30(4):3193–3223, 2024a. doi: 10.3150/23-BEJ1711.
- Matias D. Cattaneo, Michael Jansson, and Xinwei Ma. Local regression distribution estimators. *Journal of Econometrics*, 240(2):105074, 2024b. doi: 10.1016/j.jeconom.2021.01.006.
- Abhishek Chakraborty and Tianxi Cai. Efficient and adaptive linear regression in semi-supervised settings. *The Annals of Statistics*, 46(4):1541–1572, 2018. doi: 10.1214/17-AOS1594.
- Jianmin Chen, Huiyuan Wang, Thomas Lumley, Xiaowu Dai, and Yong Chen. Surrogate-powered inference: Regularization and adaptivity. *arXiv preprint arXiv:2512.21826*, 2025. doi: 10.48550/arXiv.2512.21826.
- Siyi Deng, Yang Ning, Jiwei Zhao, and Heping Zhang. Optimal and safe estimation for high-dimensional semi-supervised learning. *Journal of the American Statistical Association*, 119(548): 2748–2759, 2024. doi: 10.1080/01621459.2023.2277409.
- Wenlong Ji, Lihua Lei, and Tijana Zrnic. Predictions as surrogates: Revisiting surrogate outcomes in the age of AI. *arXiv preprint arXiv:2501.09731*, 2025. doi: 10.48550/arXiv.2501.09731.
- Nathan Kallus and Xiaojie Mao. On the role of surrogates in the efficient estimation of treatment effects with limited outcome data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):480–509, 2025. doi: 10.1093/jrssb/qkae099.
- Hyung-Chul Lee, Yoonsang Park, Soo Bin Yoon, Seong Mi Yang, Dongnyeok Park, and Chul-Woo Jung. VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. *Scientific Data*, 9:279, 2022. doi: 10.1038/s41597-022-01411-5.
- Sida Li and Nikolaos Ignatiadis. Prediction-powered adaptive shrinkage estimation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of*

- Machine Learning Research*, pages 34836–34875. PMLR, 2025. URL <https://proceedings.mlr.press/v267/li25ak.html>.
- Jiacheng Miao, Xinran Miao, Yixuan Wu, Jiwei Zhao, and Qiongshi Lu. Assumption-lean and data-adaptive post-prediction inference. *Journal of Machine Learning Research*, 26(179):1–31, 2025. URL <https://jmlr.org/papers/v26/24-0056.html>.
- Margaret Sullivan Pepe. Inference using surrogate outcome data and a validation sample. *Biometrika*, 79(2):355–365, 1992. doi: 10.1093/biomet/79.2.355.
- James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994. doi: 10.1080/01621459.1994.10476818.
- Shanshan Song, Yuanyuan Lin, and Yong Zhou. A general M-estimation theory in semi-supervised framework. *Journal of the American Statistical Association*, 119(546):1065–1075, 2024. doi: 10.1080/01621459.2023.2169699.
- Yang Sui, Jin Zhou, Hua Zhou, and Xiaowu Dai. Prediction-powered conditional inference. *arXiv preprint arXiv:2603.05575*, 2026. doi: 10.48550/arXiv.2603.05575.
- Shota Takeishi, Jimin Ding, and Xuming He. On prediction-powered inference for quantile regression via convolution smoothing. *arXiv preprint arXiv:2606.04128*, 2026. doi: 10.48550/arXiv.2606.04128.
- Ran Tao, Donglin Zeng, and Dan-Yu Lin. Efficient semiparametric inference under two-phase sampling, with applications to genetic association studies. *Journal of the American Statistical Association*, 112(520):1468–1476, 2017. doi: 10.1080/01621459.2017.1295864.
- Lorenzo Testa, Qi Xu, Jing Lei, and Kathryn Roeder. Semiparametric semi-supervised learning for general targets under distribution shift and decaying overlap. *arXiv preprint arXiv:2505.06452*, 2025. doi: 10.48550/arXiv.2505.06452.

- Anastasios A. Tsiatis. *Semiparametric Theory and Missing Data*. Springer Series in Statistics. Springer, New York, 2006. doi: 10.1007/0-387-37345-4.
- U.S. Environmental Protection Agency Office of Research and Development. Dataset long term performance of six air sensors across the united states. U.S. EPA ScienceHub data set, 2025. URL <https://catalog.data.gov/dataset/dataset-long-term-performance-of-six-air-sensors-across-the-united-states>.
- Pengyun Wang. Efficient transported distributional and quantile treatment effects with surrogate-assisted missing primary outcomes. *arXiv preprint arXiv:2605.02154*, 2026. doi: 10.48550/arXiv.2605.02154.
- Anru Zhang, Lawrence D. Brown, and T. Tony Cai. Semi-supervised inference: General theory and estimation of means. *The Annals of Statistics*, 47(5):2538–2566, 2019. doi: 10.1214/18-AOS1756.
- Tijana Zrnic and Emmanuel J. Candès. Cross-prediction-powered inference. *Proceedings of the National Academy of Sciences*, 121(15):e2322083121, 2024. doi: 10.1073/pnas.2322083121.